Farmer Conekt Poster

Ishami Irené | Ineza kevin| Nganji Jospin| Mukabucyana Dedine | Mukamukiza Milliam

African Leadership University | BSc (Hons) Software Engineering | Research + Demo Poster

1. Problem, Context, and Users

In spite of this significance perishable crops like fruits and vegetables caused by limited connection with dependable markets, inadequate coordination between sellers and purchasers, and the legacy of middlemen who tend to control costs, are estimated at 20-40% (MINAGRI, 2022). Without great timing from the farmer getting to buyer the loss keep going despite agriculture contributing 25 percent of the national GDP.

- **What broken**: slow timing for harvest to be sold due to many middlemen.
- **Who is most affected**: Farmers, who losses when harvest unsold
- **Why is this worth solving**: protecting farmers from loss and consumer from overpriced products.

2. Objectives and Scope

- **Aim:** Build and maintain Farmer conekt system an e-commerce for farmer to sell their products as fast as possible with one middleman
- **Objective 1:** Design System architecture featuring harvested, pre harvested and post harvested Products.
- **Objective 2:** Implement the Minimum Viable Product(MVP) with payment, listings, and registration.
- **Objective 3:** Test it's working on 50 farmers, and 50 buyers around Masoro and Kimironko market.
- **Scope:** includes registration, product list, buying and paying; excludes forecasting.

3. Project Metadata

Team	Ishami Irené, Ineza Kevin, Mukabucyana Dedine, Mukamukiza Milliam, Nganji Jospin
Programme	BSc (Hons) Software Engineering / Cohort 3
Supervisors	African Leadership University (ALU)
Demo / QR	**Scan for Demo**

For OTP on payment validation: `123456` since, we used testing version

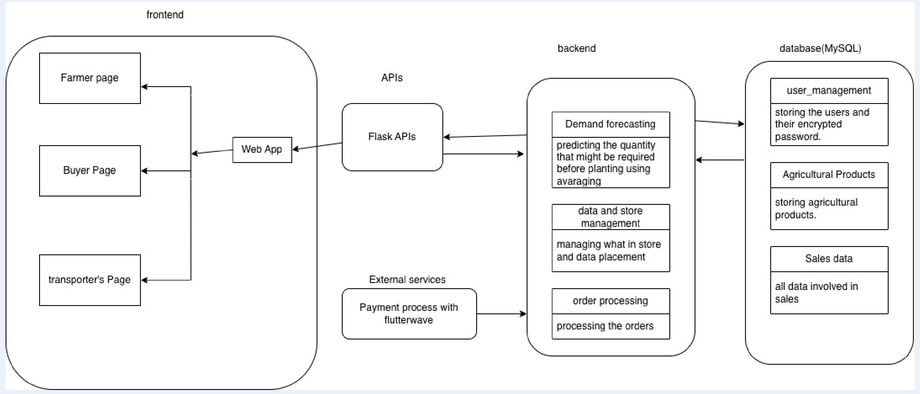
4. Constraints and Design Decisions

- **Data quality and privacy**: Login is required before any data is sent to the server, like orders. We also implemented encryption and used MySQL to ensure secure storage and data integrity.
- **Mode stack and reason**: HTML/CSS, vanilla JavaScript (frontend), Python with Flask (backend/APIs), and MySQL (database).
- **Main trade-off**: We optimized for simplicity and collaboration, sacrificing scalability and advanced performance features.

5. Proposed Solution and System Overview

Solution summary
Farmers upload their products either before or after harvesting. Buyers browse, place orders, and make payments. Orders are assigned to transporters who deliver products to the destination through a unified system.

Architecture checklist
- frontend
- Backend
- APIs
- Database



6. Methodology and Build Process

Research / discovery
- Research conducted via articles and papers
- Feasibility validated through local market context
- Risks include disputes after delivery if mismatch occurs

Engineering / implementation
- Architecture and design choices
- Prototype iterations and testing
- Deployment, simulation, or demo setup

7. Results, Evaluation, and Demo Evidence

- **Navigation Reliability:** 100% success rate in end-to-end user flow, from registration to product listing.
- **System Responsiveness:** Latency for real-time market updates.
- **Data Integrity:** Successful between the frontend and the backend with authentication.
- **Demo URL:** https://farmers-conekt.ineza.tech

100%
Navigation success

8. Impact, Contribution, and Learning

- Technical contribution: designed system where any interaction with server need login and browser products without any interaction with server.
- User benefit: consumer get fresh products and with fair price, and farmer get buyers as fast as possible.
- Research insights: keeping people to learn about these technology will continue since there is a big gap between youth and agriculture.
- Mission relevance: As ALU is mission driven study, there is 30–50% of post harvest lost in journey to final consumer, with this technology and learning is going to reduce the number as low as possible.

9. Limitations and Next Steps

- **Limitations in testing and scale**: Trust remains a major challenge, as some users are reluctant to adopt online payments beyond familiar mobile money services. There is also hesitation to pay for goods before delivery or to dispatch products without upfront payment. Additionally, low levels of computer literacy among many farmers may hinder effective adoption and use of the system
- **Known assumptions and constraints**: The system assumes that users have access to the internet and are able to use the platform effectively. It also relies on users being honest in describing their products and providing accurate images.
- **Future improvements and scaling**: Future work includes integrating product demand forecasting and introducing trust or verification badges to identify reliable farmers, thereby increasing user confidence and strengthening trust within the platform.

10. References and Acknowledgements

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